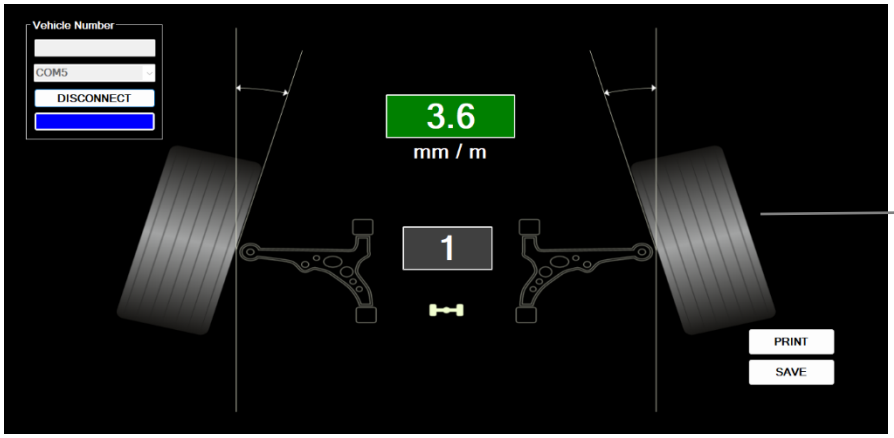
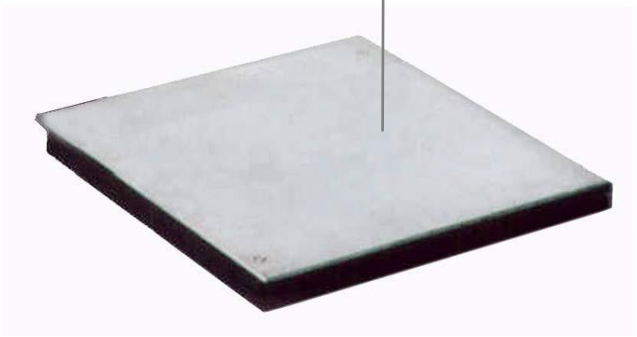
SSP 2000 / 2500 / 4000

Operating Manual Side Slip Tester

Ref.-No. 2000601966



Digital Display
with PC Software



mechanics
track plate

CARTEC®

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1. Introduction

1.1 About this Manual

We congratulate you on your decision to buy a ***CARTEC* Side-Slip Tester,** series **SSP 2000 / SSP 2500 / SSP 4000.**

You have acquired a high-grade testing unit.

Prior to dispatch, each individual test unit has passed various tests to ensure

THIS CARTEC SIDE-SLIP TESTER FUNCTIONS PERFECTLY!

In order to exclude possible trouble with the equipment, please take note of the following indications:

- Keep the unit dry!
- Protect the unit from intense solar radiation!
- Connect the unit to the appropriate PC system!
- Put the column on solid and even floor!



IMPORTANT:

- **Only authorized personnel is allowed to carry out initial operation and repair work!**
- **Improper handling or sabotage entails the waiver of warranty claims!**
- **Make sure that electric repair is exclusively carried out by electricians or authorized personnel!**
- **Only trained personnel is allowed to use and operate the Side-Slip Tester!**

In case of any unexpected defect, please inform the service team in charge of your workshop. They will be pleased to be at your service.

1.2 Abbreviations

Please note the explanations of the abbreviations used in CARTEC-Operating Manuals, as well as in Service & Maintenance Instructions and in Installation Instructions.

Products:

BDE	= Roller Brake Tester
PROFI	= Plate Brake Tester
FWT	= Suspension Tester
SSP	= Sideslip Tester
GST	= Joint Play Detector

General Terms:

VDA	= front axle
HIA	= rear axle
HDB	= hand brake
FSB	= parking brake
BBA	= service brake system
NFZ	= commercial vehicle

1.3 Manufacturer´s Data

Product Use

Side-Slip Testers - series **SSP 2000 / SSP 2500 / SSP 4000** allow easy and quick performance of side-slip tests with passenger cars, commercial vehicles, busses, and trucks. That applies to vehicles equipped with single axle drive.The test stand is particularly designed for the quick diagnosis in workshops.

Easy operation connected with a quick test run allows time-saving and efficient “in-line tests“ after a short training period.

Manufacturer

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2. Initial Operation

- Connect the unit to the PC system using USB.
- Open software then the unit adjusts itself to zero now.

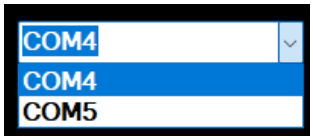
Incorrect vehicle wheel alignments cause irregular and excessive tire wear. The roadworthiness is correspondingly reduced, and the vehicle is exposed to avoidable overstress, as for example to wheel bearings, track joints and the like. Wheel alignment by visual means is time and cost intensive.

Wheel alignment by the **SSP 2000 / SSP 2500** or **SSP 4000** is quickly and easily per- formed. In no time you obtain an indication whether the vehicle requires visualwheel alignment. There is no risk of a wheel alignment on spec. At the same time, you strengthen your customer´s confidence in his service station as no superfluous and cost intensive repairs at customer´s expense is necessary using **SSP 2000 / SSP2500** or **SSP 4000.**

The application of the **SSP 2000 / SSP 2500** or **SSP 4000** is worthwhile for you and your customer and serves to increase roadworthiness.

3. Test Sequence

Open the SSP software 

Select communication port 

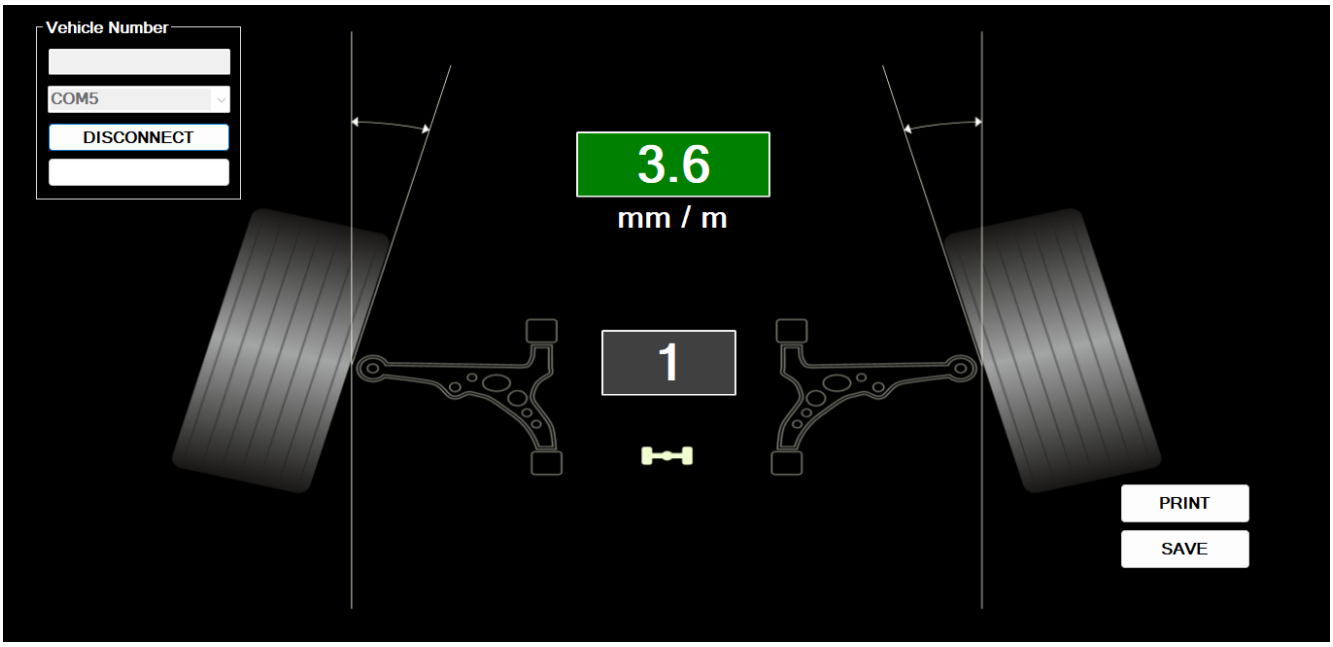
Click CONNECT button 

After connected to the unit indicator must be blinking 

Drive the vehicle to be tested straight over the test plate. The test speed should range between 3 and 5 km/h.

Stop as soon as the front wheel has completely rolled over the plate.

Read now the track deviation in "*mm/m*" on the display. The axle symbol toe-out (-) or toe-in (+).



For test sequence side slip tester SSP combined with brake tester see corresponding operating manuals.

3.1 Measuring Errors

- Zero indication not correctly set despite of moving the track plate.
- Keep steering wheel steady while driving over the track plate!



IMPORTANT:
Do not move the steering wheel!

- It is important that the test plate is completely passed. The wheels must leave the test plate.
- Drive the vehicle straight over the test plate. Driving over the plate in diagonal direction falsifies the measured result.
- Check the correct connection of the digital indicator ±. Please refer to INITIAL OPERATION:
- As an option, it is possible to store the measured and obtain a printout by measure of an A4 printer.

4. Fault Finding

Should the **SSP 2000 / SSP 2500** or **SSP 4000** unexpectedly not be ready for operation, please check the following points: (Please refer to Connection Plan.)

4.1 Mains Voltage

Measure the voltage of USB data cable at the mains 5V USB. Check USB data cable.

4.2 Low Voltage

If the electronic board does not turn on, it is inevitably necessary to have the entire electronic board repaired.

4.3 Track Potentiometer

Check whether the electric resistance between terminal “1“ and “3“ is 2K Ohm. Additionally check whether between terminal “1“ and “2“ the electric resistance can be changed between 0 Ohm. 2 K Ohm by shifting the track plate.



IMPORTANT:
Disconnect the potentiometer for this measurement. Renew the track potentiometer (displacement sensor), if the prescribed resis- tance values are not reached.

5. Electric Connection

5.1 General

The **Side-Slip Tester** is supplied ex works with the connecting cable between track plate and display. However, the track plate cable is already pre-installed in the track plate.

For the connection to the local electrical PC, there is a supply lead UBS with data cable required.



IMPORTANT:
Use the SSP 2000 / SSP 2500 / SSP 4000 in the following way:

a) track plate with display cabinet on its own

b) track plate in combination with Brake Tester or Suspension Tester.

5.2 Electric connection with own display cabinet

The control lead of the track plate is labelled with the numbers 1, 2, and 3 in the free cable ends. Again, connect them according to diagram.

- cable no. 1 …white (GND)
- cable no. 2 …brown (signal)
- cable no. 3 …green (+Vcc)

5.3 Electric connection as a combination unit without own display

For this see, the following terminal connection diagrams:

BDE 2204 / 2304
VIDEO*line*
BDE 3504 / 4504
4xx BR

The test results are given via the installed display or PC monitor.

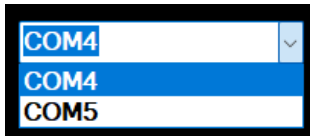
6. Calibration

6.1 Track Calibration Proceedings: single unit

- Open the SSP software



- Select communication port



- Click CONNECT button



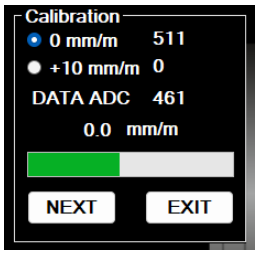
- After connected to the unit indicator must be blinking



- Press "Ctrl" and "F1" simultaneously

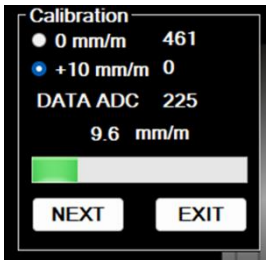
- Open calibration in menu

- Calibration box will be open

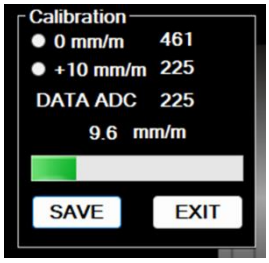


- Do not move the track plate during this procedure since the actual track zero point is determined this way then click NEXT

- To exactly 10mm right deflect the track plate and have the calibration value then click NEXT



- To save calibration value click SAVE



6.2 Track Calibration Proceedings: combined device

See operating manuals of corresponding brake testers fort this.

7. Technical Data

general		
axle weight **SSP 2000 / 2500**	(t)	4
axle weight **SSP 4000**	(t)	18
temperature range	(°C)	from 0 to 70
software unit		
dimension	(pixel)	1280x720
display		digital
electronics		
microprocessor		digital
principle of measurement		potentiometer
mechanics Side-Slip Plate		
length x width x height **SSP 2000**	(mm)	1000 x 600 x 50
weight	(kg)	70
length x width x height **SSP 2500**	(mm)	500 x 570 x 50
weight	(kg)	20
length x width x height **SSP 4000**	(mm)	1000 x 800 x 71
weight	(kg)	153

8. Installation Instructions

1. Compare pit dimensions with drawing and make sure that the pit is free of dirt. Lay the empty pipe from the test plate to the display cabinet. Make sure that the empty pipe has no sharp curves and is free of dirt and mounting residues.

2. Position the unpacked equipment on two pieces of wood (outside the pit).

3. Undo the screws and remove top plate.

4. Remove the cables from the frame.

5. Lift the complete frame into the clean and prepared pit.



ATTENTION: Avoid damaging the cables!

6. Lead the cables into the switch cabinet.

7. Now set the mechanic part into the pit and adjust it so that it is exactly placed in drive direction in the center of the pit. Fix the mechanic part at plan height. After mounting the top plate, the upper edge of the track plate must be equal with the workshop floor. The frame of the mechanics must have a square fit over the entire pit floor. There should be not rocking movement.

8. Bolt down the frame.

9. connect the cable of the mechanics to the display unit/switch box.

10. Loosen the M10 transportation nuts. Check the proper functioning of the mechanics and the potentiometer transducer. It must go 100% to mid-position.

11. Set the zero point and re-check the functions. (If in connection with a line system, then this is done automatically.)

12. Check the side slip results on the display. The polarity sign gives the toe-in (+) or toe-out (-). If necessary, please change the connection in the cabinet.

13. If everything is o.k., mount the top plate, tighten the screws and re-check the functions. Attention SSP 2500: The correct mounting of the cover plate is very important; the plate must hook into the lifting safety device: lifting control.

14. Please carry out the work thoroughly to exclude later complaints, instruct the customer accordingly and hand over the equipment.

9. Maintenance Instructions - Mechanics

Guide Roller, Ball Bearing Traveler and Bolt Guide Side-Slip Tester SSP 2000 / 2500 / 4000



NOTE:
During the maintenance work, it is important to pay attention to the effective safety regulations!

- The most economic regreasing period can only be determined according to the specific operating conditions of each individual test station. The calculated regreasing period amounts to approximately 1000 operating hours or, in case of an interrupted operation mode, once a year.

- Remove cover plates.

- Clean guide rollers, ball bearing travelers and bolt guides with pressure springs from the previously soiled layer of grease. The whole blank sliding surfaces of the existing guidance's must be greased with ball bearing grease of mean consistence.

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EC-Declaration of Conformity according to the EC Standards for Machines 89/392/EEC Annex II A

Herewith we declare that the machine, as indicated below, with regard to the general performance and construction corresponds with the fundamental safety and health requirements stipulated in the EC Standards for Machines.

Any change made in the machine without the approval of Snap-on will deprive this Declaration of its validity.

Designation of the machine :	**Side Slip Tester**
Typ:	**SSP 2000**
Corresponding EC Standards:	EC Standards MACHINES 98/37/EEC, 98/79/EEC EMC directive 89/336/EEC, 92/31/EEC, 93/68/EEC Low voltage directive 73/23/EEC, 93/68/EEC
Especially applied harmonised standards:	EN 60204 - 1 EN 50081 - 1 : 1992 EN 50082 - 1 : 1992

Date: 01.12.2006

Klaus Kirstätter
Director

CARTEC

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Any change made in the machine without the approval of Snap-on will deprive this Declaration of its validity.

Designation of the machine :	**Side Slip Tester**
Typ:	**SSP 2500**
Corresponding EC Standards:	EC Standards MACHINES 98/37/EEC, 98/79/EEC EMC directive 89/336/EEC, 92/31/EEC, 93/68/EEC Low voltage directive 73/23/EEC, 93/68/EEC
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Designation of the machine :	**Side Slip Tester**
Typ:	**SSP 4000**
Corresponding EC Standards:	EC Standards MACHINES 98/37/EEC, 98/79/EEC EMC directive 89/336/EEC, 92/31/EEC, 93/68/EEC Low voltage directive 73/23/EEC, 93/68/EEC
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